

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. (EE) Examination

December 2012

EE405 Power System Protection

Date: 07.12.2012, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Choose the correct option.

[07]

- I. When the back-up protection operates, there is..
(a) a wider disruption than when only primary protection operates
(b) lesser disruption
(c) equal disruption in case of both
(d) no disruption
- II. The number of input quantities and settings for a directional relay is:
(a) a single relaying quantity and a single setting
(b) two relaying quantities and two settings
(c) two relaying quantities and one setting
(d) a single relaying quantity and two settings
- III. The operating time of an IDMT relay corresponding to $PSM=1$ and $TMS=1$ will be:
(a) 1 s , (b) 3 s , (c) infinite , (d) 30 s
- IV. Which of the voltage-current combinations corresponds to the 30° connection:
(a) I_a, V_{ac} , (b) I_a, V_{ca} , (c) $I_a, V_{ac} + V_{bc}$, (d) $I_a, V_{ac} + V_{cb}$
- V. As seen by differential protection of a transformer, the inrush condition appears like:
(a) external fault, (b) internal fault, (c) through fault, (d) slight overload
- VI. The main purpose of “ Interposing CT” used in transformer differential protection is:
(a) fine tuning the ratio of transformation
(b) compensation for phase angle error
(c) isolation from high voltage
(d) compensation of phase shift produced by delta-star or star-delta transformer
- VII. Impedance relay is:
(a) over-current relay with voltage restraint
(b) over-current relay with directional restraint
(c) directional relay with voltage restraint
(d) directional relay with current restraint

Q - 2 (a) CHARUSAT is going to install a 1250 KVA, 11KV/415V, three phase distribution transformer at medical campus. Though it is 11KV class equipment, due to its high rating it is necessary to protect the same transformer by biased differential relay. The relevant details as per following. **[08]**

- (i) Transformer:
 - OLTC: -7.5 % to +10% in steps of 2.5 % of rated voltage on H.V. side.
 - Vector Group: DY11, Rated frequency 50 Hz, Neutral solid grounded.
 - No load current is 3% of the rated current
- (ii) Current Transformers:
 - L.V. side : 2000/5 A star Connected
 - H.V. side : 75/1 A star Connected
 - Interposing C.T.S. connected on H.V. side
 - C.T.S. with Ratio Error = ± 3 %
- (iii) Biased Differential Relay :
 - Relay Rating : 5 Amp
 - Basic setting: 5 % to 25 % of rating (in steps of 5 %)
 - Bias setting: (10 - 50 % in steps of 10 %)
- (iv) Air break Circuit Breaker:
 - Rated Voltage: 415 Volts
 - Breaking Capacity: 10 MVA

Find out ICT ratio and draw detailed three phase wiring diagram of a transformer differential protection scheme. Also suggest suitable settings of the relay.

(b) Complete the sentences by one or more phrase. **[03]**

- I. E-shaped or shaded pole core is used in IDMT over current relay to....
- II. Dead zone for directional relay can be decrease by...
- III. Discriminative time between two successive coordinated IDMT relays used for transmission line protection can be decrease by....

(c) Give Reason: For the protection of transmission line, IDMT relays are preferred over the inverse time-current relay. **[03]**

OR

Q - 2 (a) Figure 1 (Drawn on page no. 5) shows a single line diagram of a part of power system. Relay used are IDMT over current relays and rating of relays are as per CT secondary rating. Setting range of relay is 50-200% of rated current in step of 25%. Determine the plug setting of all relays and TMS of R_1 , R_2 , R_3 if TMS of R_4 and R_5 are set to 0.1 and 0.15 respectively. Also the high set instantaneous setting of relays R_3 and R_5 are set to 1400% of the CT secondary rating. **[08]**

(b) Give reason: over fluxing protection is provided in case of generator transformer. **[02]**

(c) Give Reason: In case of distance relay, the first zone setting is restricted by transient where as the third zone setting is restricted by possible over load. **[02]**

(d) With neat vector diagrams, explain why we are using 90° connection directional relay where the fault current is lagging by almost 90° . **[02]**

- Q - 3 (a)** Figure 2 (Drawn on page no. 5) shows a portion of a power system network. Relay R is a mho distance relay having a MTA=70°. Find three zone setting if [08]
- (a) Zone I covers 85% of the line L_2
 - (b) Zone II reaches up to bus C
 - (c) Zone III covers maximum 50% line length of 220 KV transmission line.
- (b)** With neat diagram explain the blocking carrier scheme used with carrier aided distance protection. [06]

OR

- Q - 3 (a)** Figure 3 (Drawn on page no. 6) shows a single line diagram of a part of power system. Relay used are directional or non-directional IDMT over current relays of 1 amp rating. Setting range of relay is 50-200% of rated current in step of 25%. Determine which relays required directional features and find the plug setting and TMS of all relays, if TMS of R_9 is set to 0.15. [08]
- (b)** Give Reason: Restricted earth fault protection is required in large power transformer, even if we are providing differential protection to the transformer. [02]
- (c)** Name two conditions where distance relay overreaches and name two conditions where it under reaches. [02]
- (d)** Why it is necessary to compensate phase shift for three phase transformer differential protection. [02]

SECTION – II

- Q - 4 (a)** Explain following terms in brief: [04]
- I. Unit & Non unit protection.
 - II. Relay Characteristics & Thermal ability curve
- (b)** Write down the special superior features of numerical digital relay which are absent in electro mechanical version. [03]
- Q - 5 (a)** List out all protection applied to Induction motor. [04]
- (b)** Draw neat diagram & Explain Frame leakage protection of Busbar in brief. [04]
- (c)** With polarity marking on both side of CT, draw differential protection scheme for generator. [04]
- (d)** Define & explain the following terms for current transformer [02]
- I. Accuracy limiting factor
 - II. Knee point voltage
 - III. Rated accuracy limit primary current
 - IV. Rated short circuit current

OR

- Q - 5 (a)** Draw & Explain 'The Duplicate busbar Protection'. Explain the operation for at least one location of fault. Assume one feeder as incoming & remaining one as outgoing. [10]
- (b)** Explain Negative Phase sequence Relay with necessary diagrams. [04]

- Q - 6 (a)** Explain the current injection technique (Primary injection test) to test simple three phase OC + E/F relay? Explain the same test for checking CT Ratio, Polarity, tightness of CT secondary wiring, Pick up current & time of operation of relay. [04]
- (b)** Explain the problems present in estimation of phasor at time of fault. Derive equation for estimation of unknown voltage phasor using two samples per data window. [10]

OR

- Q - 6 (a)** solve the numerical: [09]

Generator Specification:

100 MVA, 11 KV, 3 Phase, 50 Hz, Star connected Alternator

Differential Protection Datas:

CT Ratio: 3000/5

Relay Pick up setting: 150 milli Amp

Relay Resistance: 100 ohms

Resistance of wire: 0.54 ohms per 100 meters.

Distance between two set of CT: 250 meters.

For the fault current of 14 times the full load current, the emf developed in one CT is 85 % of the emf developed in other ended CT.

- I. Find out the location where relay would have zero voltage across itself.
 - II. If Relay is to be putted at physical midpoint, find out value of stabilization resistance for stability factor of 3.
 - III. Find out the distance up to which the relay may be connected without any stabilization resistance. Assume stability factor of one.
- (b)** Derive the equation for stabilization resistance R_{stab} required to be inserted with relay for differential protection scheme of generator. [05]




